

LAXMAN KANAGARATNAM

Department of Astrophysics, University of Vienna
laxman.kanagaratnam@univie.ac.at · laxmankanagaratnam.github.io
github.com/laxmankanagaratnam · orcid.org/0009-0006-7566-4137

EDUCATION

MSc Astronomy, University of Vienna	2026 – 2028 (expected)
BSc Physics, University of Vienna 141/180 ECTS completed Thesis advisors: Prof. Christoph Dellago and Alena Rottensteiner	2025 – 2027 (expected)
BSc Astronomy, University of Vienna Thesis advisor: Alena Rottensteiner	2024 – 2026
MIT OpenCourseWare (self-study) Physics lectures, used to compensate for course schedule conflicts	2025 – present

RESEARCH EXPERIENCE

Alves Lab, Department of Astrophysics, University of Vienna Advisor: Alena Rottensteiner (PhD student, University of Vienna)	2025 – present
• Bachelor thesis on the large-scale structure of the Vela region using Gaia DR3, applying the SIGMA clustering algorithm (developed by Sebastian Ratzenböck) to identify and characterise stellar groups. • Currently developing a noise-removal strategy for SIGMA, improving its robustness against field-star contamination.	

TEACHING EXPERIENCE

Teaching Assistant, University of Vienna Dynamics and Thermodynamics — Prof. Kristina Kislyakova and Prof. Glenn van de Ven	2026
Teaching Assistant, University of Veterinary Medicine Vienna Medical Physics Laboratory — Dr. Olga Jovanovic	2025
Teaching Assistant and Senior Coach, ETH Zürich Analysis I — Prof. Alessio Figalli and Dr. Alexander Caspar	2023

OUTREACH & SERVICE

Academic community contributions, University of Vienna Semester summaries and exam-preparation notes for BSc physics and astronomy students	2024 – present
aCentauri Solar Racing, ETH Zürich Competitor in the Bridgestone World Solar Challenge 2023	2023
Physics I formula compendium, ETH Zürich Co-developed the official $\LaTeX$ compendium — Prof. Werner Wegscheider	2023
Saturday Morning Physics, University of Basel Matura project exhibitor — PD Dr. Matthias Liebendörfer	2020

TECHNICAL SKILLS & ACTIVITIES

Programming & tools	Python (NumPy, matplotlib, pandas, Astropy), Jupyter, $\LaTeX$, Git/GitHub
Astronomy software	TOPCAT, SIMBAD, Muniwin, VStar, Minima27
Extracurricular	Private tutor for mathematics and physics; former teacher and member, Chess Club Therwil

Last updated: July 2026